Lecture 05

28.10.2025

So far: basic examples of PDEs and numerical schemes

Now: rigorous (theoretical) introduction of some specific <u>classes</u> of PDEs

<u>Definition</u> Let $\Omega \subseteq \mathbb{R}^n$ be a <u>domain</u>. An expression of the form

$$(*) \quad 0 = F(D^k u(x), D^{k-1} u(x), \dots, Du(x), u(x), x)$$

for all $x \in \Omega$, where $D^j u$, $j = 1, \dots, k$ are the partial derivatives of u of order j, and where

$$F : \mathbb{R}^{(n^k)} \times \mathbb{R}^{(n^{k-1})} \times \dots \times \mathbb{R}^n \times \mathbb{R} \times \Omega \to \mathbb{R}$$

is called a (scalar) <u>k-th order PDE</u> for the unknown $u : \Omega \to \mathbb{R}$. A function u that satisfies $(*)$ is called a <u>solution of the PDE</u>.

Example (Hamilton-Jacobi-Bellman equation)

$$\nabla u(x)^T f(x) - |\nabla u(x)^T g(x)|^2 + l(x) = 0$$

where $f: \mathbb{R}^n \to \mathbb{R}^n$, $g: \mathbb{R}^n \to \mathbb{R}^n$, $l: \mathbb{R}^n \to \mathbb{R}$.

$\rightsquigarrow$ too complicated for an introductory course

We will mainly consider a simpler class of PDEs:

Definition A k-th order PDE of the form

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$$\sum_{|\alpha| \leq k} a_\alpha(x) D^\alpha u(x) = f(x)$$

with given coefficient functions $a_\alpha: \Omega \to \mathbb{R}$ and right-hand side $f: \Omega \to \mathbb{R}$ is called **linear**. For $f = 0$, the PDE is called **homogeneous** otherwise **inhomogeneous**.

Remark Note that linear PDEs satisfy a so-called **superposition principle**, i.e., if

u_1, u_2 are solutions to

$$\sum_{|\alpha| \le k} a_\alpha(x) D^\alpha u_i(x) = f_i(x) \quad , i = 1, 2,$$

then $\tilde{u} := \underline{u_1 + 2u_2}$ is a solution to

$$\sum_{|\alpha| \le k} a_\alpha(x) D^\alpha \tilde{u}(x) = \underline{f_1(x) + 2 f_2(x)}$$

Even more special case: $k = 2$, $n = 2$ $\rightsquigarrow$ e.g. $|\alpha| = 2$

corresponds to $D^{(0,2)} u(x)$, $D^{(1,1)} u(x)$, $D^{(2,0)} u(x)$

so $\dfrac{\partial^2 u}{\partial x_2^2}$, $\dfrac{\partial^2 u}{\partial x_1 \partial x_2}$, $\dfrac{\partial^2 u}{\partial x_1^2}$

or, more generally:

<u>Definition</u> Let $\Omega \subset \mathbb{R}^n$ be a domain. A <u>linear second-order PDE</u> for the unknown $u: \Omega \to \mathbb{R}$ is given by

$$-\sum_{i,k=1}^{n} a_{ik}(x) \underbrace{u_{x_i x_k}}_{\text{or: } \frac{\partial^2 u}{\partial x_i \partial x_k}}(x) + \sum_{i=1}^{n} b_i(x) u_{x_i}(x)$$

$$+ \quad c(x)\, u(x) = f(x) \qquad \forall\, x \in \Omega$$

where $a_{ik}, b_i, c, f : \Omega \to \mathbb{R}$ are given functions for $i, k = 1, \dots, n$. If a_{ik}, b_i, c are independent of $x \in \Omega$, we say that the PDE has <u>constant coefficients</u>, otherwise <u>variable coefficients</u>. If $f(x) = 0$ for all $x \in \Omega$, we call the PDE <u>homogeneous</u>, else <u>inhomogeneous</u>. If one variable is interpreted as "time", we call the PDE <u>instationary</u>, else <u>stationary</u>.

<u>Remark</u> if u is two times cont. differentiable, then

$$u_{x_i x_k} = u_{x_k x_i}$$

$\Rightarrow$ we can therefore assume that $a_{ik}(x) = a_{ki}(x)$ and also

$$A(x) = \begin{bmatrix} a_{11}(x) & \cdots & a_{1n}(x) \\ \vdots & & \vdots \\ a_{n1}(x) & \cdots & a_{nn}(x) \end{bmatrix} = A(x)^T$$

We call A the <u>diffusion matrix</u>

Recall that a symmetric matrix $A \in \mathbb{R}^{n \times n}$ has n real eigenvalues. We can therefore define:

<u>Definition</u> (classification of linear 2nd order PDEs)

A linear, second-order PDE is called

(a) <u>elliptic</u> in $x \in \Omega$ if $A(x)$ is definite, i.e., all eigenvalues are either strictly positive or strictly negative.

(b) <u>hyperbolic</u> in $x \in \Omega$ if $A(x)$ has one strictly negative eigenvalue and $n-1$ strictly positive eigenvalues (or vice versa).

(c) <u>parabolic</u> in $x \in \Omega$ if $A(x)$ has one eigenvalue zero and $n-1$ strictly positive (or strictly negative) eigenvalues and $\mathrm{rk}([A(x), b(x)]) = n$, where $b(x) = \begin{bmatrix} b_1(x) \\ \vdots \\ b_n(x) \end{bmatrix}$.

The PDE is called elliptic / hyperbolic / parabolic if it is elliptic / hyperbolic / parabolic in all $x \in \Omega$.

If none of the above cases is true, we call the PDE <u>unclassified</u>.

<u>Examples</u>

(a) Poisson equation:

$$-\Delta u(x) = -u_{x_1x_1}(x) - \dots - u_{x_nx_n}(x) = f(x)$$

$$A = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{pmatrix} \rightarrow \text{only positive eigenvalues}$$

$\rightarrow$ linear, second order PDE, elliptic, stationary, inhomogeneous

(b) Heat equation (in d spatial coordinates $z_1, \dots, z_d$)

$$u_t(t,z) = \Delta u(t,z) + f(t,z), \quad \text{here: } \Delta u(t,z) := u_{z_1z_1} + u_{z_2z_2} + \dots + u_{z_dz_d}$$

With regard to the definition, we have $n := 1+d$.

$b_1 = 1, \; b_i = 0, \; i = 2, \dots, \underbrace{d+1}_{=n}$

$$\underset{\mathbb{R}^{(d+1)\times(d+1)}}{A} = \begin{pmatrix} 0 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{pmatrix} \rightarrow \text{eigenvalues } 0, 1 \; (d = n-1 \text{ times})$$

$$\operatorname{rk}([A, b]) = \operatorname{rk}\left(\left[\begin{array}{cccc|c} 0 & & & & 1 \\ & 1 & & & 0 \\ & & \ddots & & \vdots \\ & & & 1 & 0 \end{array}\right]\right) = d+1 = n$$

$\rightarrow$ linear, second order PDE, parabolic, instationary, inhomogeneous

(c) Wave equation (in d spatial coordinates)

$$\underline{u_{tt}(t,z)} - \underline{\Delta u(t,z)} = f(t,z)$$

Here: $\Delta u = u_{z_1 z_1} + u_{z_2 z_2} + \dots + u_{z_d z_d}$

$$\rightarrow A = \begin{pmatrix} \underline{-1} & & & 0 \\ & \underline{1} & & \\ & & \underline{1} & \\ 0 & & & \ddots \; \underline{1} \end{pmatrix}$$

one strictly negative, $\underbrace{d}_{=n-1}$ strictly positive evs

linear, second order PDE, hyperbolic, instationary, inhomogeneous